Ria Mary Suresh



Master's student in Data Science (AI/ML specialization) at FAU Erlangen-Nürnberg with a completed Bachelor's in Computer Science (GPA 1.5) and proven hands-on experience developing LLM-based applications, AI agents, and data processing pipelines. Proficient in Python, LangChain, and LangGraph. Experienced in building production-ready AI solutions from prototype to deployment, with clean code, clear documentation, and collaborative development practices.

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Ria Mary Suresh

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Education

- **Masters in Data Science (AI/ML Specialisation)** *Oct 2025 - Present*
  Friedrich-Alexander-Universität (FAU) Erlangen-Nürnberg
  Relevant Coursework: Knowledge Discovery in Databases, Deep Learning, Pattern Recognition, Decision Theory, AI in Medical Robotics
- **BTech in Computer Science and Engineering** *Sept 2021 - June 2025*
  Vellore Institute of Technology, Vellore, India
  Relevant Coursework: Natural Language Processing, Big Data, Data Structures and Algorithms, Software Engineering, Operating Systems
  GPA: 9.18/10 (Bavarian equivalent: 1.5)

Work Experience

Bosch Global Software Technologies | AI Intern
Jan 2025 – May 2025

Bangalore, India

- Scenario Based Video Clipping: Developed a deep learning-based video segmentation pipeline using Python and computer vision techniques, achieving 92% accuracy in classifying scene-level patterns from continuous industrial footage.
- Extracted and structured object attributes from frame-level captions using NLP, transforming unstructured data into analyzable datasets to support tracking and reporting.
- Engineered high-performance multiprocessing pipelines for parallel image processing across multiple deep learning models, reducing processing time by 75% — critical for real-time industrial image analysis at scale.

Bosch Global Software Technologies | Data Science Intern
Aug 2023 – Oct 2023

Bangalore, India

- Explored sensor data processing and object detection algorithms in the ADAS domain, evaluating both classical and neural network-based approaches for real-time detection tasks.
- Had hands-on experience in model selection, training, and optimization and analyzed diverse datasets to extract crucial information for enhancing autonomous driving models' accuracy and performance.
- Developed an interactive Streamlit dashboard to visualise model outputs and test results with dynamic filters, enabling clear analysis and structured reporting of experimental findings.

Projects

RAG Document Q&A Assistant

Tech stack: Python, LangChain, ChromaDB, Sentence-Transformers, Groq, Streamlit, PyPDF
Built a retrieval-augmented generation (RAG) application using LangChain, ChromaDB, and Sentence-Transformers, enabling users to upload PDF documents and ask natural-language questions. Implemented document chunking, semantic embeddings, vector similarity search, and grounded LLM responses using retrieved document context and source pages. Implemented conversation memory to support multi-turn Q&A sessions, enabling contextual follow-up questions within the same document.

PawGuide AI

Tech stack: Python, TensorFlow, Scikit-learn, Streamlit, LangChain, Google Gemini API, Groq API, Transfer Learning, Computer Vision, Prompt Engineering
Designed and built a complete LangChain AI agent using Llama 3.1 — implementing tool-calling architecture, and prompt engineering pipeline. Deployed via Streamlit. Demonstrates hands-on experience with AI agent development, LLM integration, and API-based data source connectivity.

Heart Health Companion

Tech stack: Python, Streamlit, NumPy, Google Gemini API, RapidAPI, Edamam
Built a neural-network-based heart disease prediction system with an accuracy of 94.6% using the UCI dataset and deployed it via Streamlit. Integrated exercise and nutrition recommendations using APIs for a complete personalized health tool.

Facial Recognition

Tech stack: Python, TensorFlow, Keras, Scikit-learn, NumPy, Matplotlib, CNN, Computer Vision
Designed a facial recognition system using a custom CNN to identify individuals based on facial features, enabling personalized user interactions achieving 95.43% identification accuracy.

Skills

Programming: *Python, SQL, R, Java, C/C++*
Machine Learning & AI: *Scikit-Learn, TensorFlow, Deep Learning, Computer Vision, NLP, OpenCV*
Data & Analytics: *Pandas, NumPy, Power BI, Matplotlib, Microsoft Excel, Data Analysis, Data Visualization*
Generative AI & RAG: *RAG, LangChain, LangGraph, LLMs, Prompt Engineering, Embeddings, Semantic Search, AI Agents, ChromaDB, FAISS*
Tools & Platforms: *Git, GitHub, Streamlit, MongoDB, Snowflake, AWS*
Soft Skills: *Problem Solving, Crossfunctional Communication, Stakeholder Presentation, Leadership*

Certifications

Amazon Web Services Cloud Practitioner
AWS (Credential ID Y96Z19BBSEEQQ4CB)
2023

Introduction to MongoDB
MongoDB (Credential ID MDB56eu3x0wfe)
2023

Complete A.I. and Machine Learning, Data Science Bootcamp
Udemy
2023

Languages

English ●●●●● *(Native)*

German ●●○○○